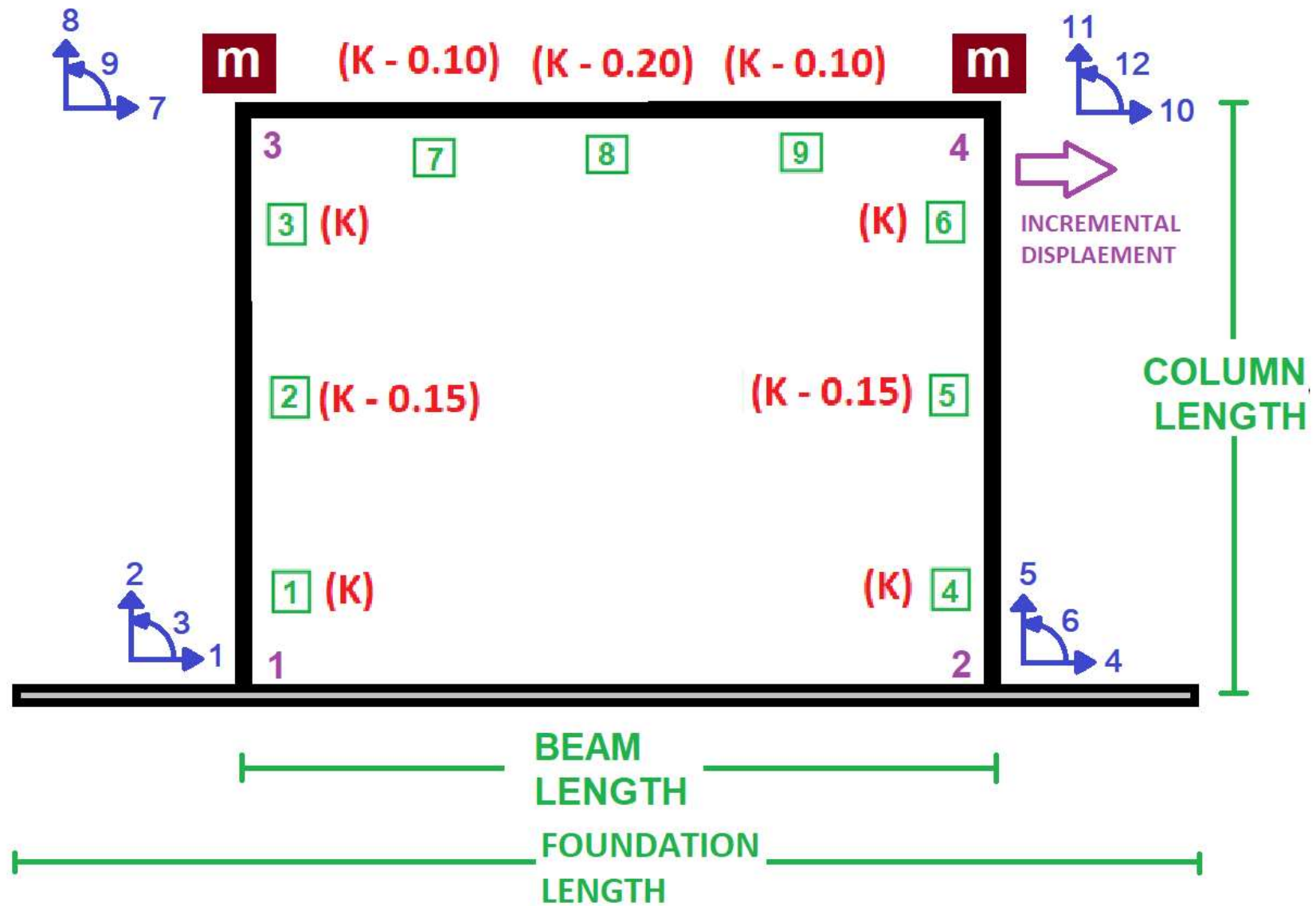
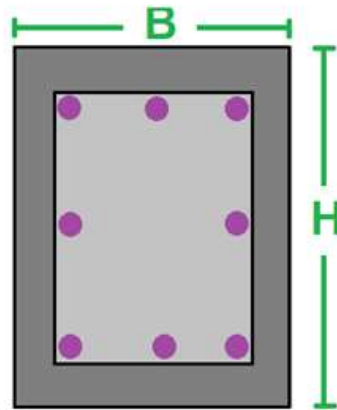


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

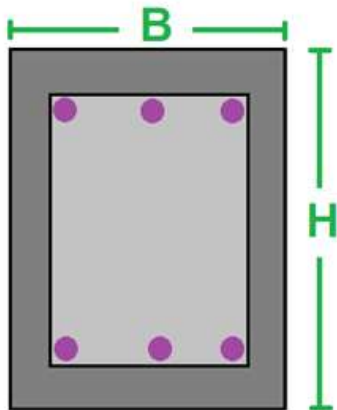
SOIL-FOUNDATION- STRUCTURE INTERACTION USING OPENSEES

SOIL SIMULATED WITH SIMPLE SPRINGS AND THE SOIL SPRINGS VALUES IS NOT EXACT.
WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

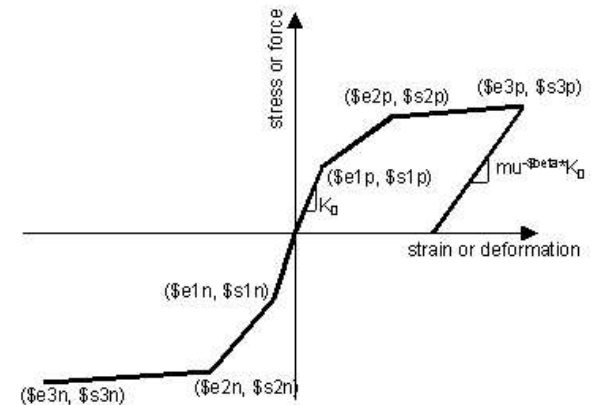
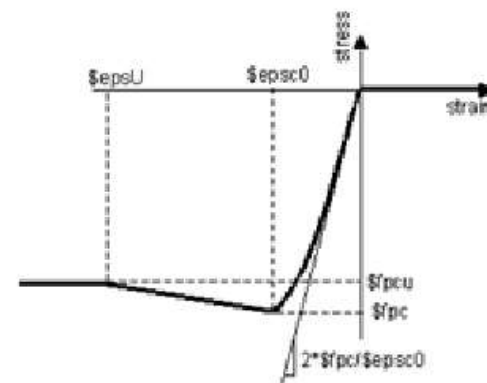
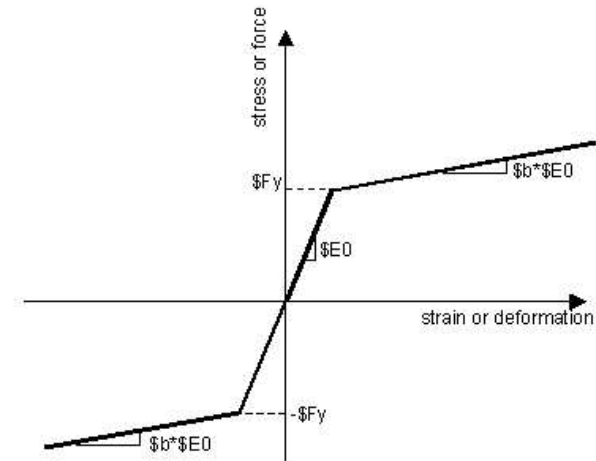
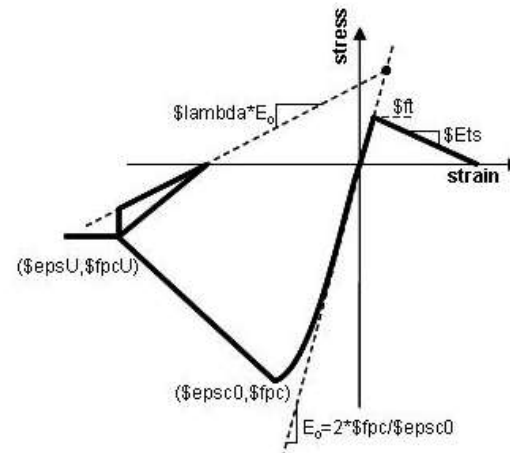


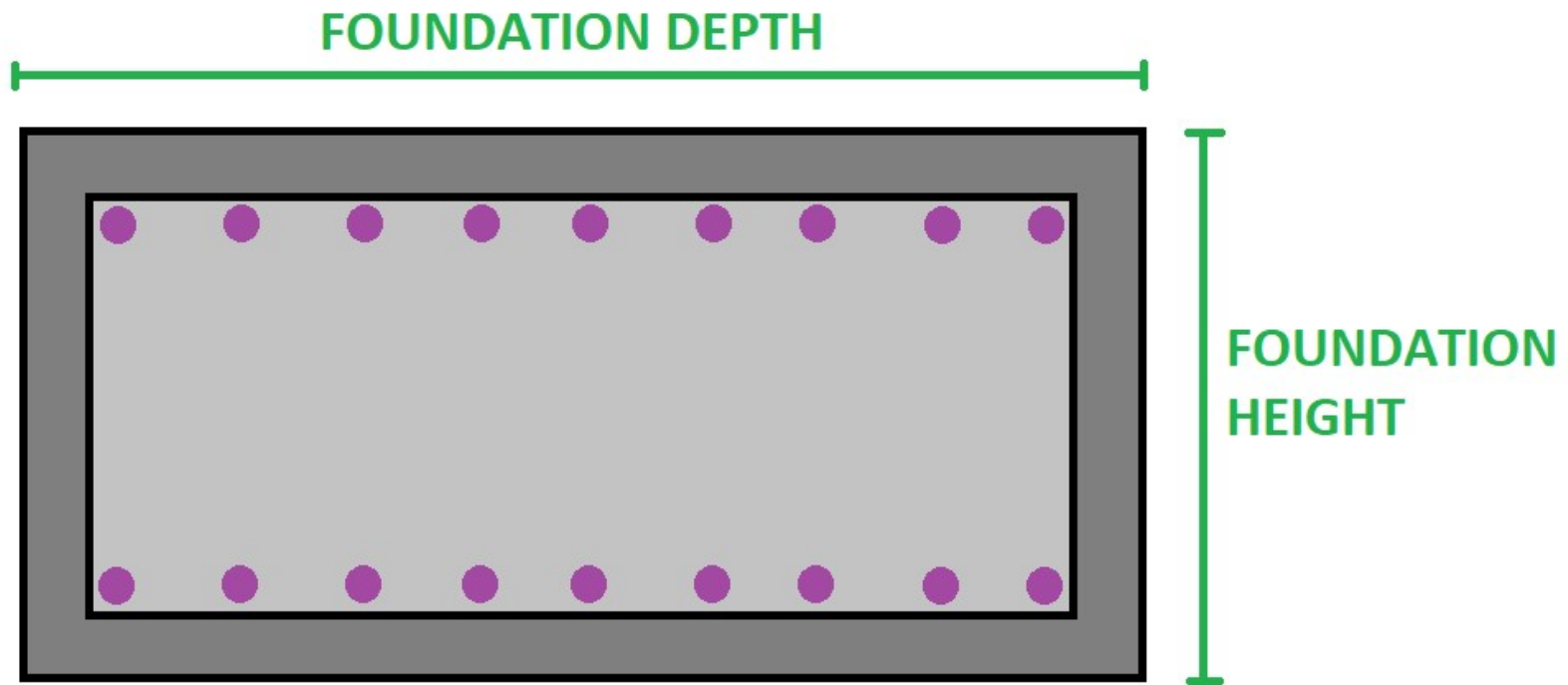


COLUMN SECTION

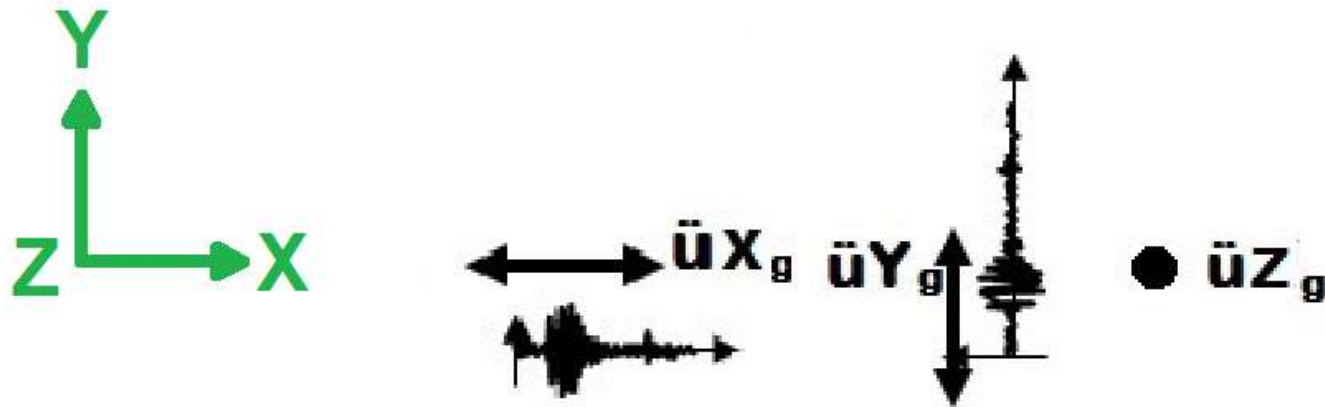


BEAM SECTION





FOUNDATION SECTION



$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

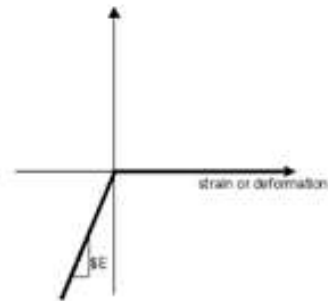
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

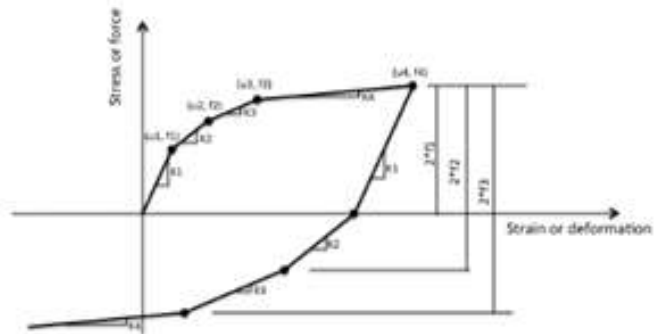
Δ_u = Lateral Ultimate Displaement from Pushover Analysis



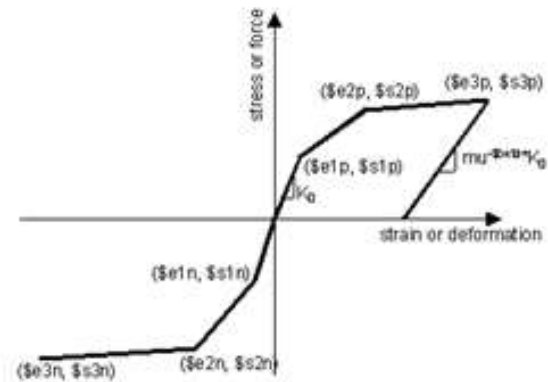
Elastic Uniaxial Material



Elastic-No Tension Material



MultiLinear Material



Hysteretic Material

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\CONCRETE_FRA...02\CONCRETE_FRAME_SOIL-FOUNDATION-STRUCTURE_02.py

CONCRETE_FRAME_SOI...ON-STRUCTURE_02.py X

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SOIL-FOUNDATION-STRUCTURE INTERACTION USING OPENSEES
4 #
5 # -----
6 # EXAMPLE 02:
7 # SOIL SIMULATED WITH SIMPLE SPRINGS AND THE SOIL SPRINGS VALUES ARE NOT EXACT.
8 # -----
9 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
10 # EMAIL: salar.d.ghashghaei@gmail.com
11 #####
12 [1] Nonlinear Frame Modeling: 2D RC frame with distributed plasticity (fiber sections) using `nonlinearBeamC
13 [2] Material Laws:
14 - *Concrete*: `Concrete01` with confined (core) and unconfined (cover) properties.
15 - *Steel*: `Hysteretic` model with pinching, hardening, and cyclic degradation.
16 [3] Seismic Loads:
17 - Pushover: Displacement-controlled lateral loading to failure.
18 - Dynamic: Uniform excitation with user-defined ground motions (X/Y components).
19 [4] Damping: Rayleigh damping (a0, a1) calibrated via eigenvalue analysis (modes 1-2).
20 [5] Performance Metrics:
21 - Ductility Ratio (μ): Derived from bilinearized pushover curves.
22 - Overstrength (Ω₀): Yield vs. ultimate capacity.
23 - Damage Index (DI): Normalized displacement demand/capacity.
24 [6] Advanced Solver: HHT-a integrator (unconditionally stable) with Newton-Raphson iterations.
25 [7] Outputs:
26 - Hysteretic responses (P-M, V-Δ, M-θ).
27 - Time-history plots (displacement, base shear).
28 - Stiffness degradation tracking.
29 [8] Validation: Logarithmic decrement method for damping ratio verification.
30 [9] Ductility Damage Index (DDI) Implementation:
31 DDI quantifies structural damage via normalized displacement demand.
32 # After bilinear fit (X[1] = Δy, X[2] = Δu):
33 Δd = max(abs(DISPL_Xd)) # Max dynamic displacement demand
34 DI = (Δd - Δy) / (Δu - Δy) # Ductility Damage Index (X-dir)
35 DI ≈ 0: Elastic response (no damage).
36 DI ≥ 1: Collapse (demand ≥ ultimate capacity).
37 Key Innovations:
38 - Combines static (pushover) and dynamic analyses for comprehensive fragility assessment.
39 - Automated bilinear curve fitting for rapid performance evaluation.
```

Console 1/A X

```
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 9(dw: << 0.00777483)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 33597.1
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 9(dw: << 0.00777483)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
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deformations for element: 9(dw: << 0.00777483)
Domain::update - domain failed in update
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OpenSees > analyze failed, returned: -2 error flag
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factor 33597.1
OpenSees > analyze failed, returned: -2 error flag
NormDispIncr KrylovNewton -2
NormDispIncr SecantNewton -2
NormDispIncr RaphsonNewton -2
NormDispIncr PeriodicNewton -2
NormDispIncr BFGS -2
```

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 317, Col 106 UTF-8 CRLF RW Mem 40%

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\CONCRETE_FRAME_ON-STRUCTURE_02\CONCRETE_SECTION_FUN_THREE.py

CONCRETE_FRAME_SOI...ON-STRUCTURE_02.py CONCRETE_SECTION_FUN_THREE.py CONCRETE_FOU

```
1 def CONFINED_CONCRETE_SECTION(secTag, h, b, cover, RD, STEEL_KIND, CONCRETE_KIND, fc, Kc, CO
2     # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAIEI (QASHQAI)
3     import openseespy.opensees as ops
4     import numpy as np
5
6     # Mass per Length
7     AREA = b * h
8     MASS = DENSITY * AREA
9
10    # Rebar Area
11    As = (np.pi * RD**2)/4
12
13    # Define parameters (units: mm, N)
14    # -----
15    # CONCRETE          tag    f'c          ec0    f'cu          ecu
16    # Parametric definitions for unconfined concrete
17    fcU = -fc          # [N/mm2] Unconfined concrete compressive stre
18    EcU = 4700 * np.sqrt(-fcU) # [N/mm2] Unconfined Concrete Elastic Modulus
19    ec0U = 2*fcU/EcU    # [mm/mm] Initial strain at peak strength (sem
20    fcUU = 0.667 * fcU  # [N/mm2] Ultimate stress (~10% of peak compre
21    ecuU = ec0U * 4.0   # [mm/mm] Ultimate strain (e.g., 3.5x ec0U)
22    lamdaU = 0.1        # Ratio between unloading slope at epsU and in
23    ftU = 0.7 * np.sqrt(-fcU) # [N/mm2] Tensile strength
24    EtsU = ftU/np.abs(ec0U) # [N/mm2] Tension softening stiffness
25
26    # Parametric definitions for confined core concrete
27    fcC = fcU * Kc      # [N/mm2] Confined strength (increased by conf
28    EcC = 4700 * np.sqrt(-fcC) # [N/mm2] Confined Concrete Elastic Modulus
29    ec0C = 2*fcC/EcC    # [mm/mm] Peak strain increases with confineme
30    fcUC = 0.6667 * fcC  # [N/mm2] Ultimate confined stress (~95% of pe
31    ecuC = 4.0 * ecuU    # [mm/mm] Ultimate confined strain (larger duc
32    lamdaC = 0.1        # Ratio between unloading slope at epsU and in
33    ftC = 0.7 * np.sqrt(-fcC) # [N/mm2] Tensile strength
34    EtsC = ftC/np.abs(ec0C) # [N/mm2] Tension softening stiffness
```

Console 1/A X

OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces & deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces & deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces & deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
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Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
NormUnbalance SecantNewton -2
NormUnbalance RaphsonNewton -2
NormUnbalance PeriodicNewton -2
NormUnbalance BFGS -2
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces & deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
NormUnbalance Broyden -2

IPython Console Files Help Variable Explorer Debugger Plots History

JupyterLab Conda: anaconda3 (Python 3.12.7) ✓ USR: Python Line 39, Col 5 UTF-8 CRLF RW Mem 40%

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\CONCRETE_FRA...N-STRUCTURE_02\CONCRETE_FOUNDATION_SECTION_FUN.py

STRUCTURE_02.py x CONCRETE_SECTION_FUN_THREE.py x CONCRETE_FOUNDATION_SECTION_FUN.py x

```
1 def CONFINED_CONCRETE_CONCRETE_FOUNDATION_SECTION_FUN(secTag, h, b, cover, RD, STEEL_KIND, C
2     # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAIEI (QASHQAI)
3     import openseespy.opensees as ops
4     import numpy as np
5
6     # Mass per Length
7     AREA = b * h
8     MASS = DENSITY * AREA
9
10    # Rebar Area
11    As = (np.pi * RD**2)/4
12
13    # Define parameters (units: mm, N)
14    # -----
15    # CONCRETE          tag    f'c          ec0    f'cu          ecu
16    # Parametric definitions for unconfined concrete
17    fcU = -fc          # [N/mm²] Unconfined concrete compressive stre
18    EcU = 4700 * np.sqrt(-fcU) # [N/mm²] Unconfined Concrete Elastic Modulus
19    ec0U = 2*fcU/EcU    # [mm/mm] Initial strain at peak strength (sem
20    fcUU = 0.667 * fcU  # [N/mm²] Ultimate stress (~10% of peak compre
21    ecuU = ec0U * 4.0   # [mm/mm] Ultimate strain (e.g., 3.5x ec0U)
22    lamdaU = 0.1        # Ratio between unloading slope at epsU and in
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24    EtsU = ftU/np.abs(ec0U) # [N/mm²] Tension softening stiffness
25
26    # Parametric definitions for confined core concrete
27    fcC = fcU * Kc      # [N/mm²] Confined strength (increased by conf
28    EcC = 4700 * np.sqrt(-fcC) # [N/mm²] Confined Concrete Elastic Modulus
29    ec0C = 2*fcC/EcC    # [mm/mm] Peak strain increases with confineme
30    fcUC = 0.6667 * fcC  # [N/mm²] Ultimate confined stress (~95% of pe
31    ecuC = 4.0 * ecuU    # [mm/mm] Ultimate confined strain (larger duc
32    lamdaC = 0.1        # Ratio between unloading slope at epsU and in
33    ftC = 0.7 * np.sqrt(-fcC) # [N/mm²] Tensile strength
34    EtsC = ftC/np.abs(ec0C) # [N/mm²] Tension softening stiffness
```

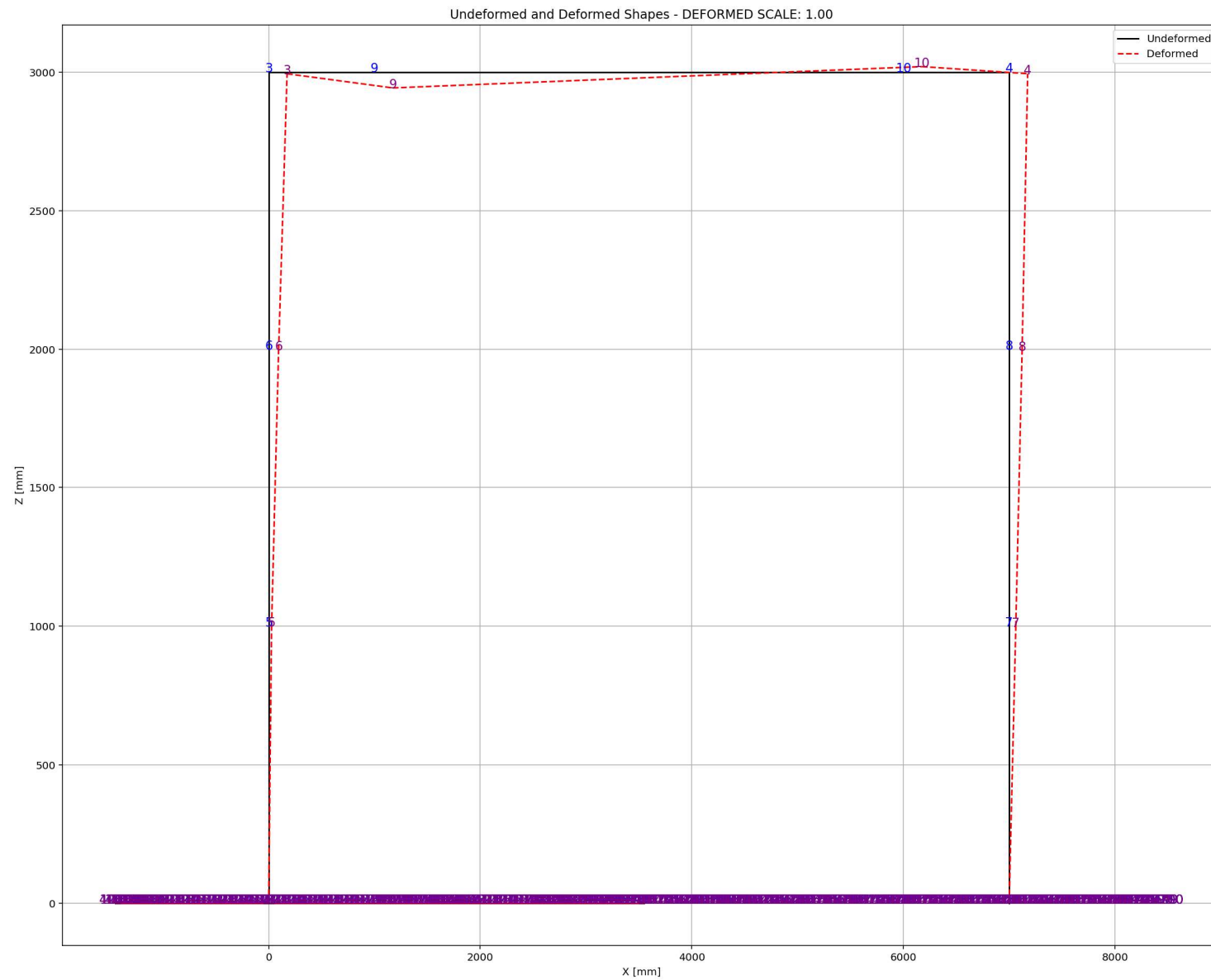
Console 1/A x

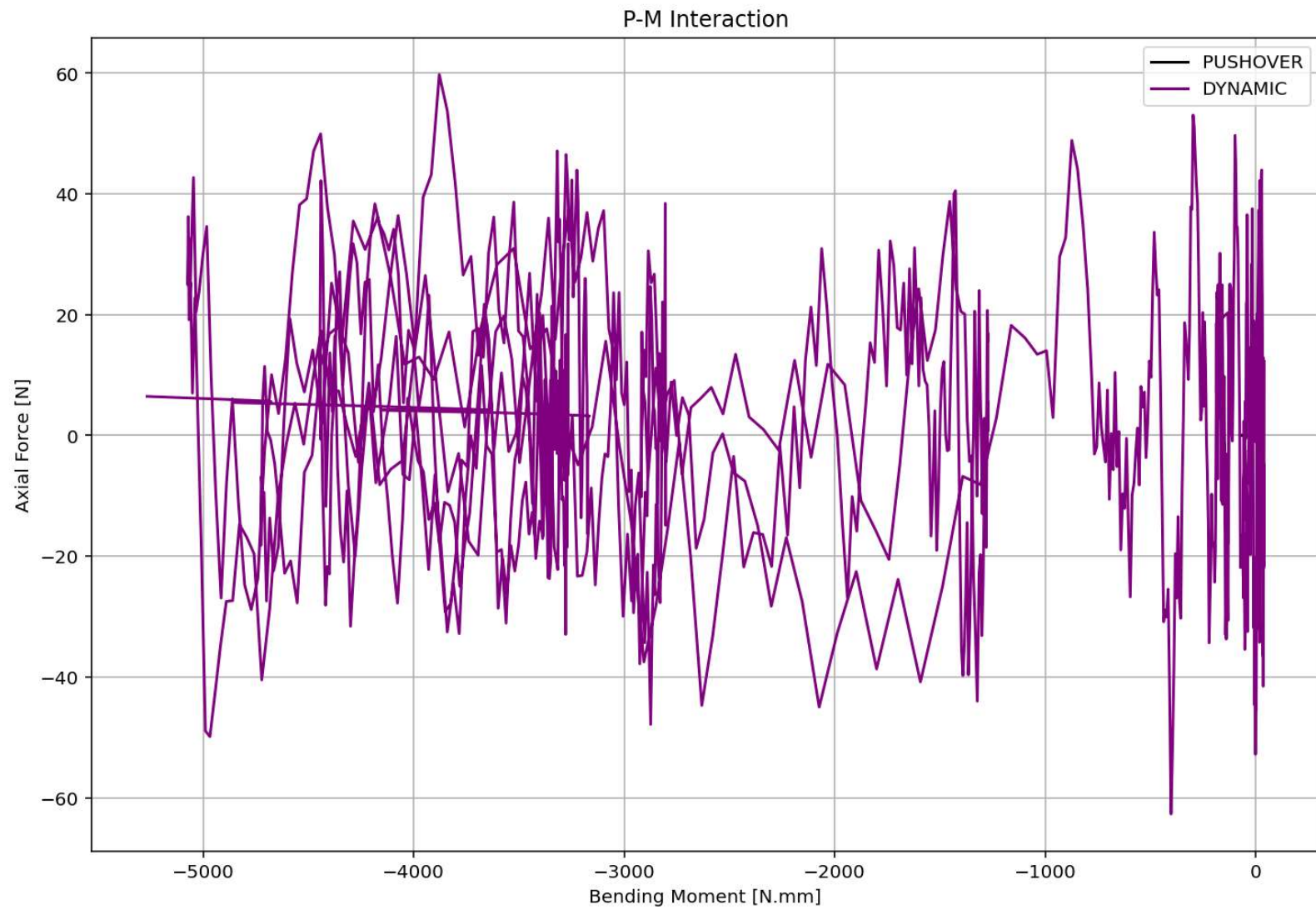
```
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 76359.5
RelativeNormUnbalance BFGS -2
RelativeNormUnbalance Broyden -2
OpenSees > analyze failed, returned: -2 error flag
WARNING - ForceBeamColumn2d::update - failed to get compatible element forces &
deformations for element: 1(dw: << 0.0124908)
Domain::update - domain failed in update
DisplacementControl::newStep - model failed to update for new dU
StaticAnalysis::analyze() - the Integrator failed at step: 0 with domain at load
factor 76359.5
OpenSees > analyze failed, returned: -2 error flag
```

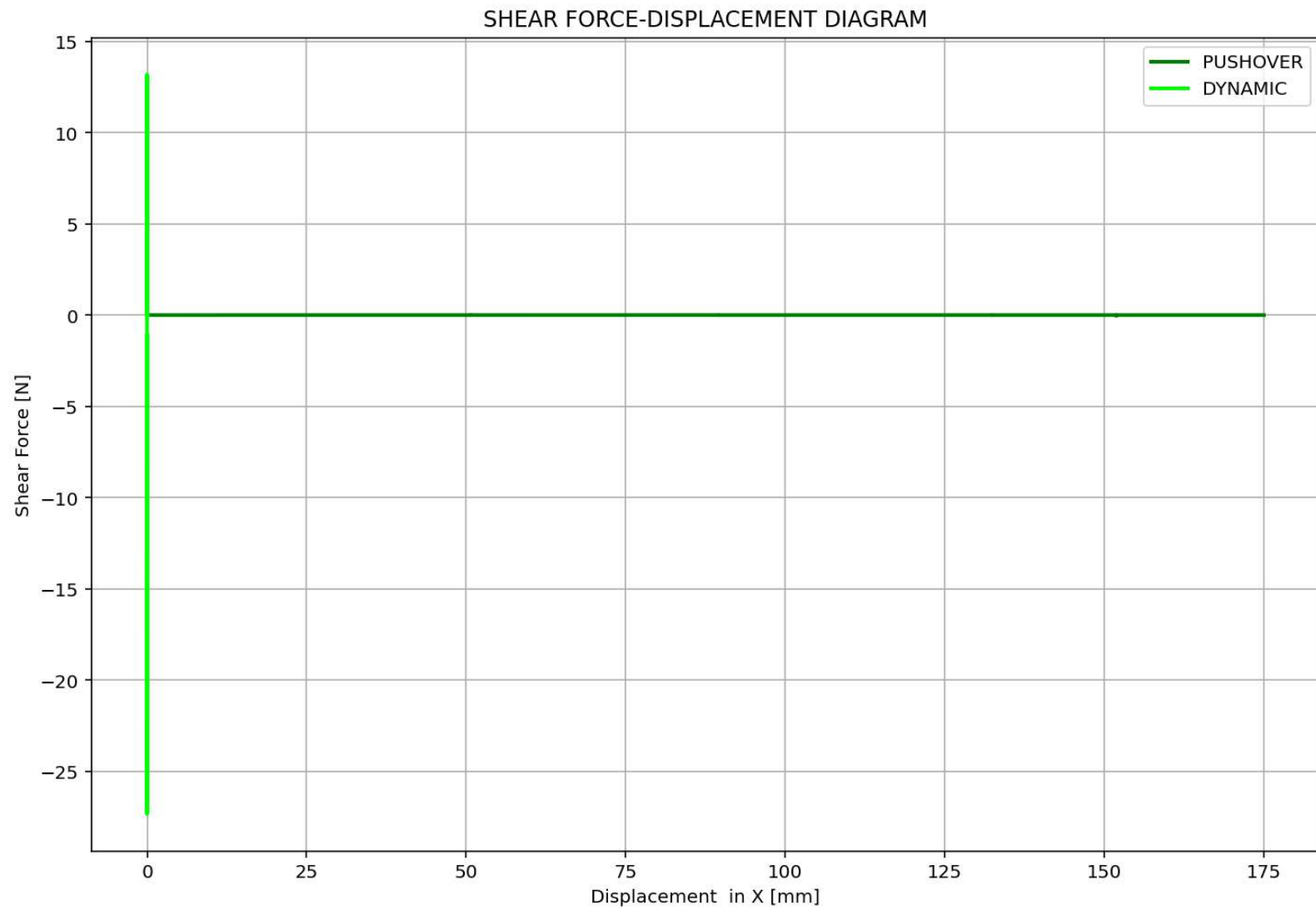
IPython Console Files Help Variable Explorer Debugger Plots History

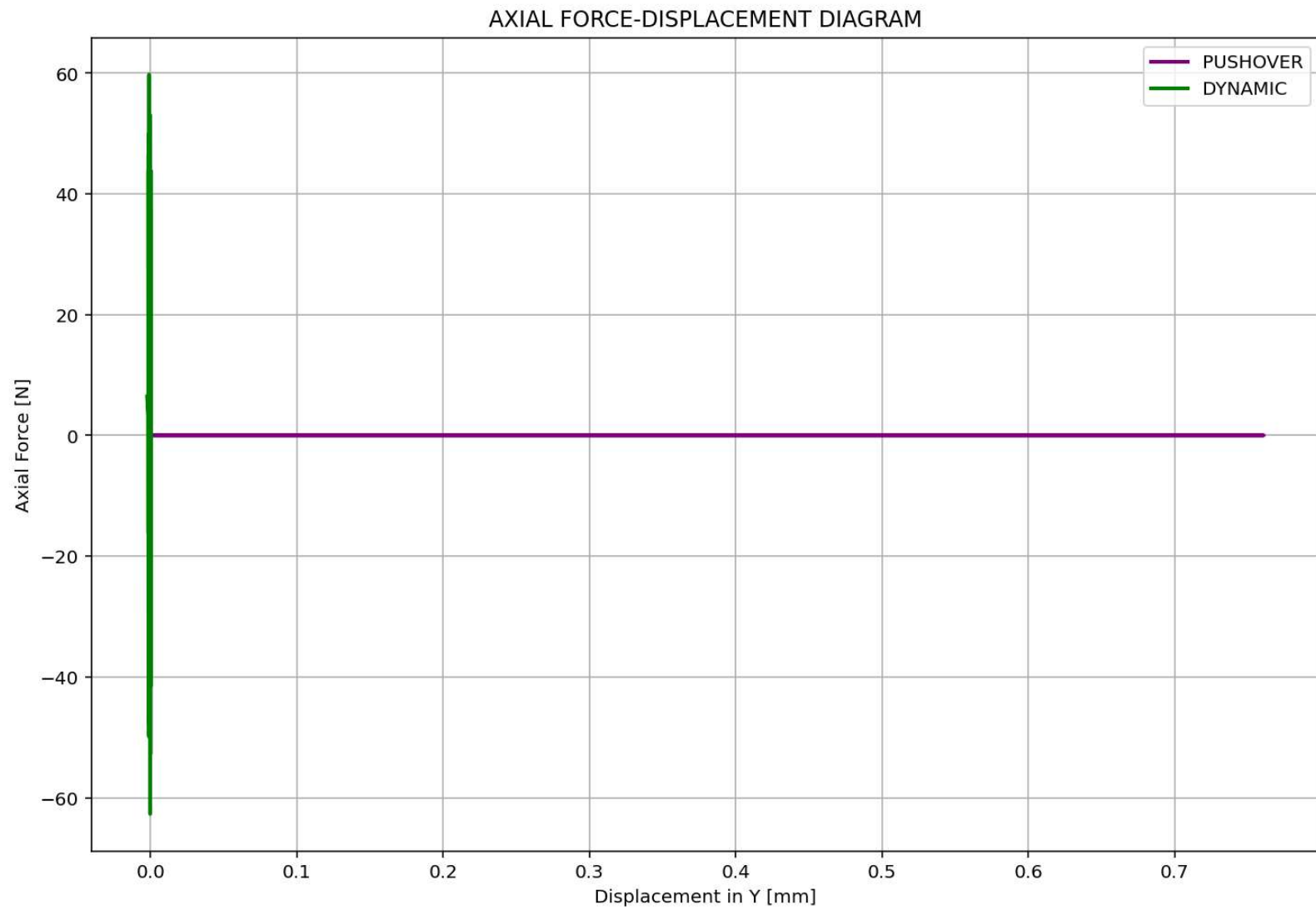
Int J. Tolmie Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 67, Col 5 UTF-8 CRLF RW Mem 41%

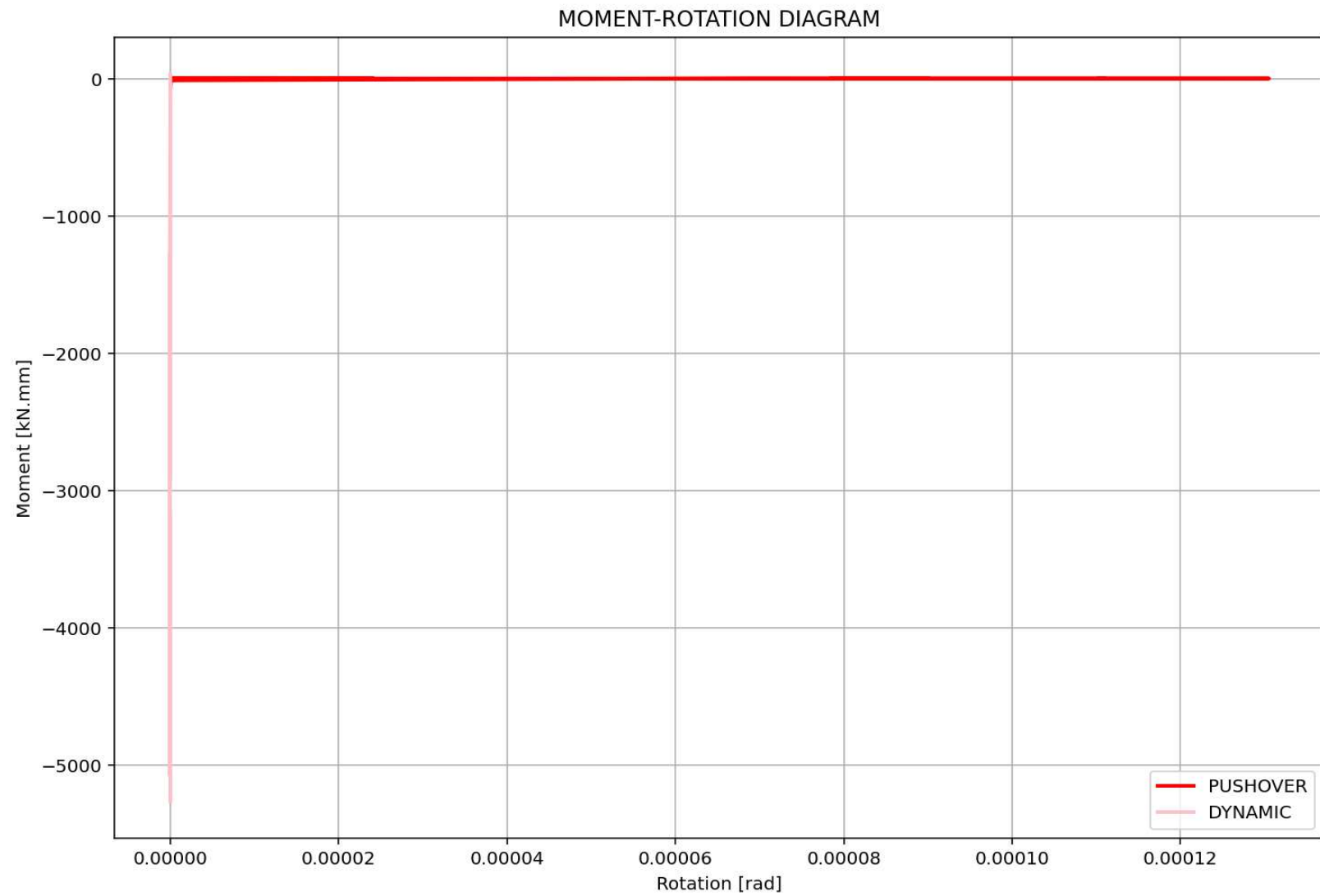
NONLINEAR STATIC ANALYSIS (PUSHOVER)



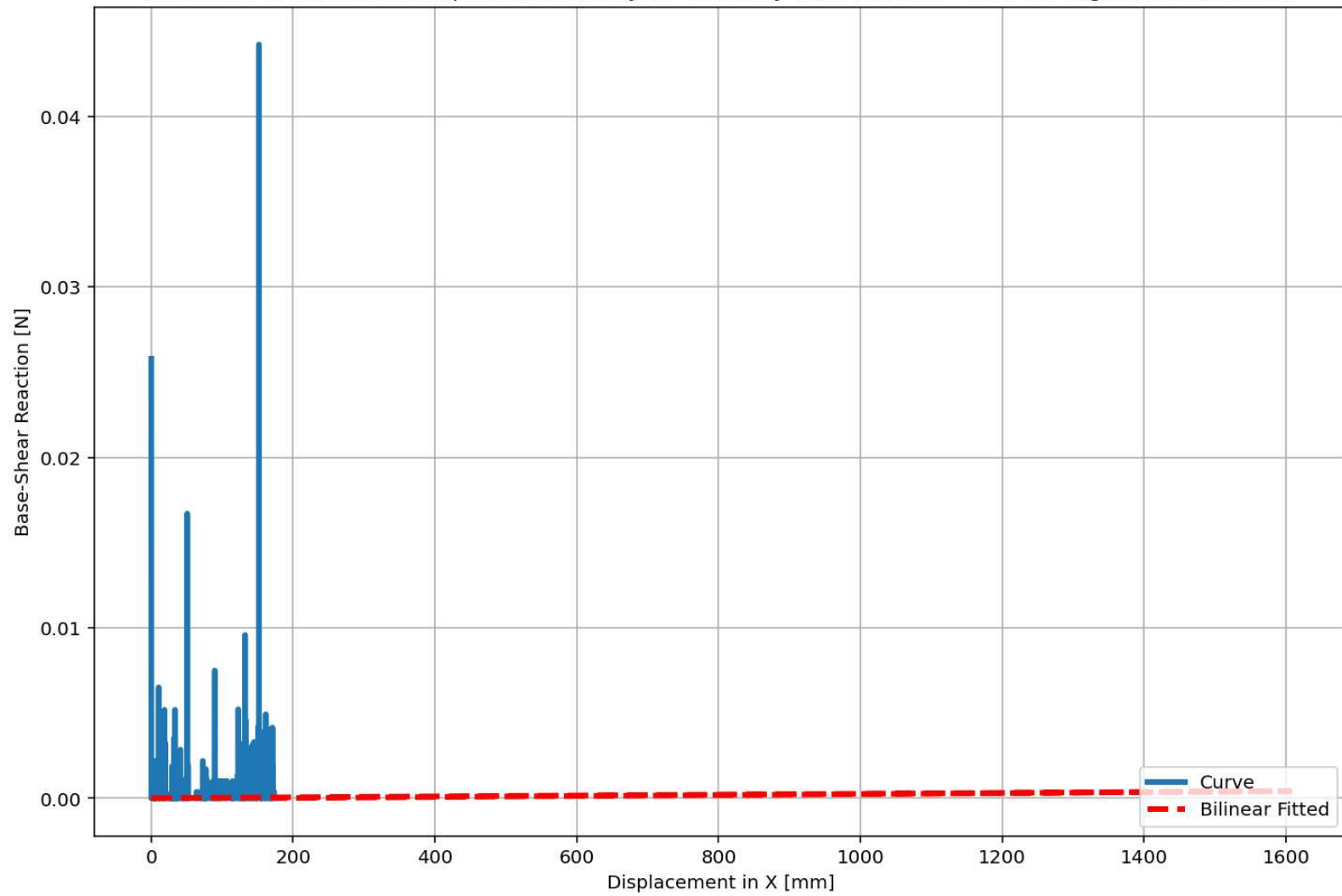






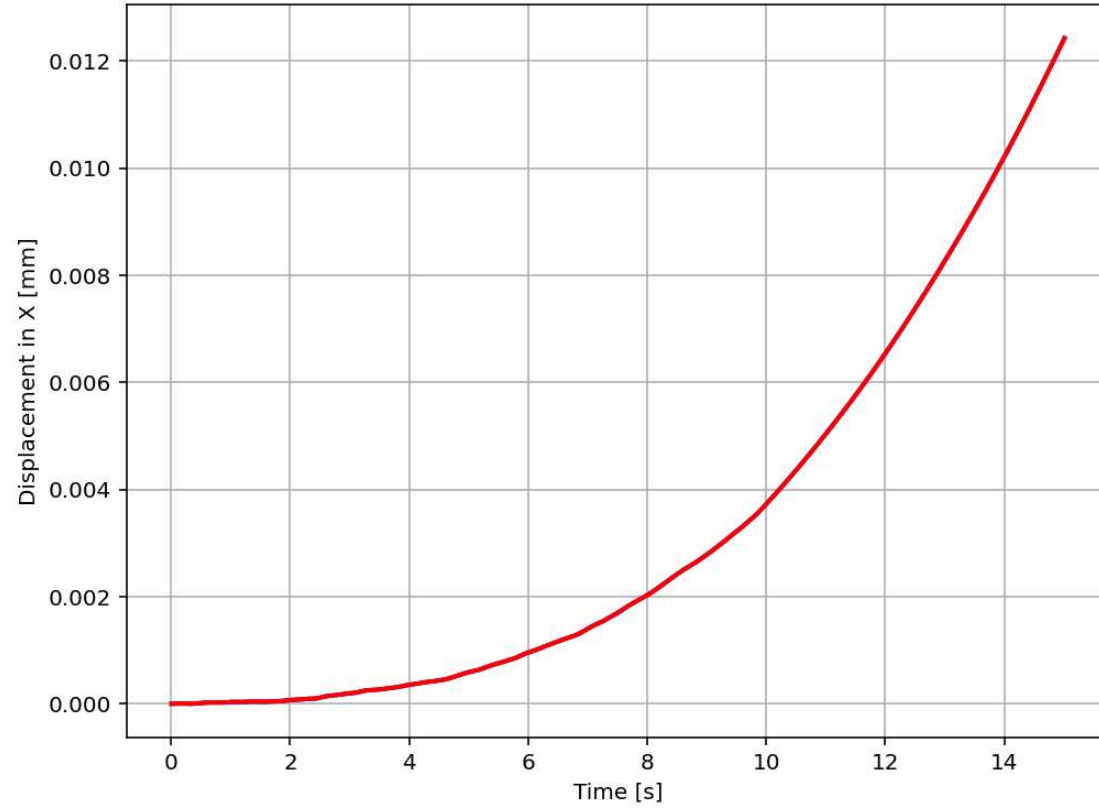


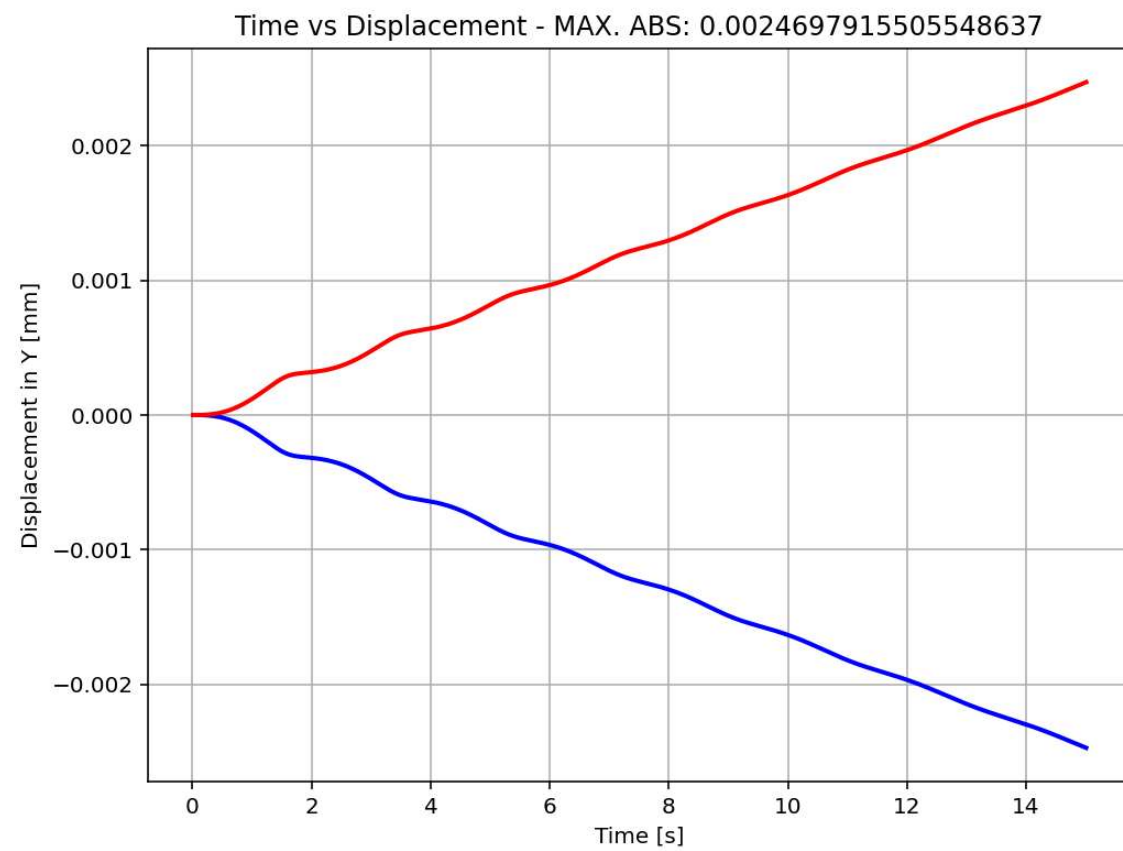
Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 0.1090 - Over Strength Factor: 0.0001

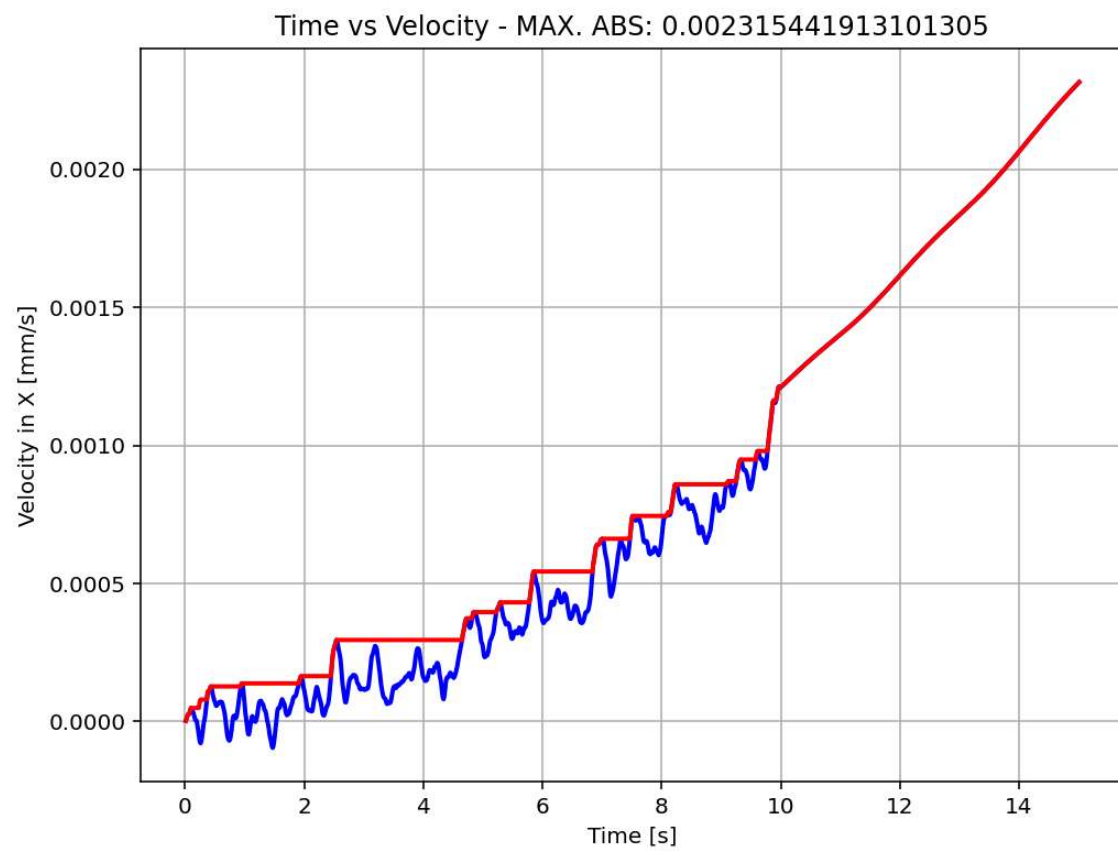


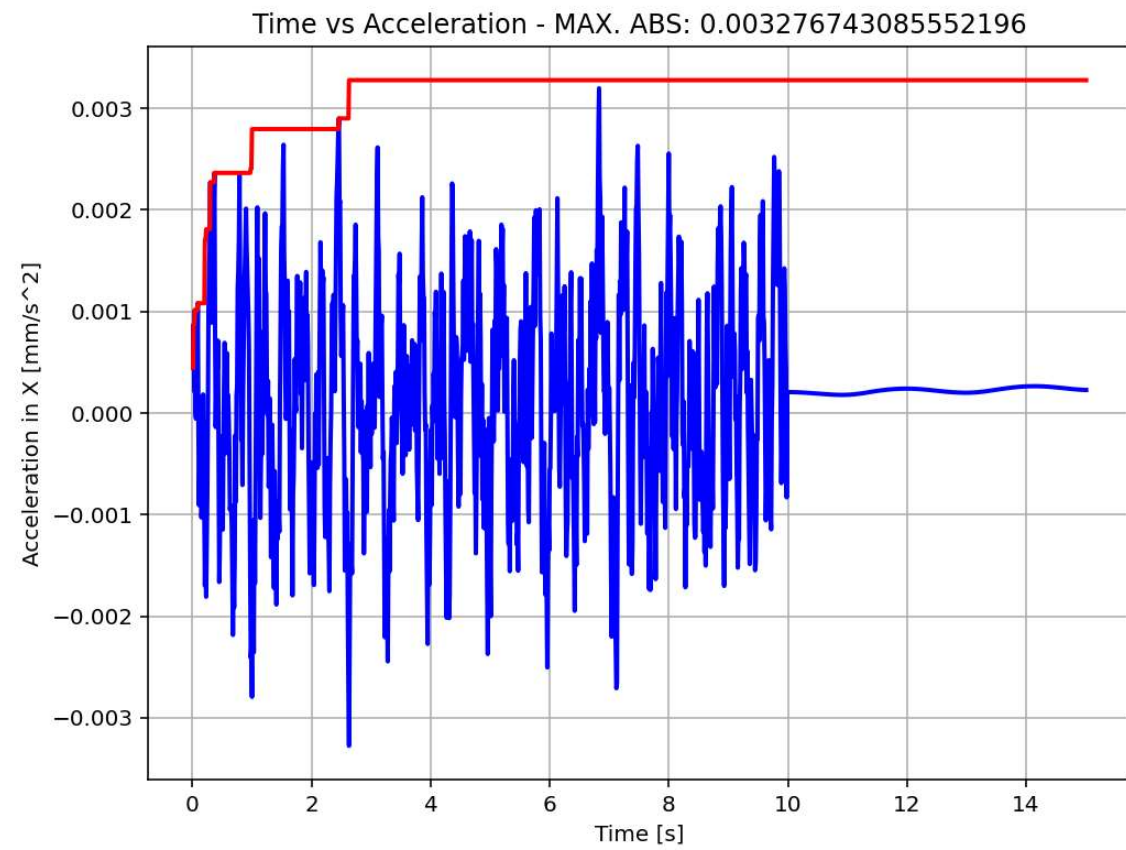
NONLINEAR DYNAMIC ANALYSIS

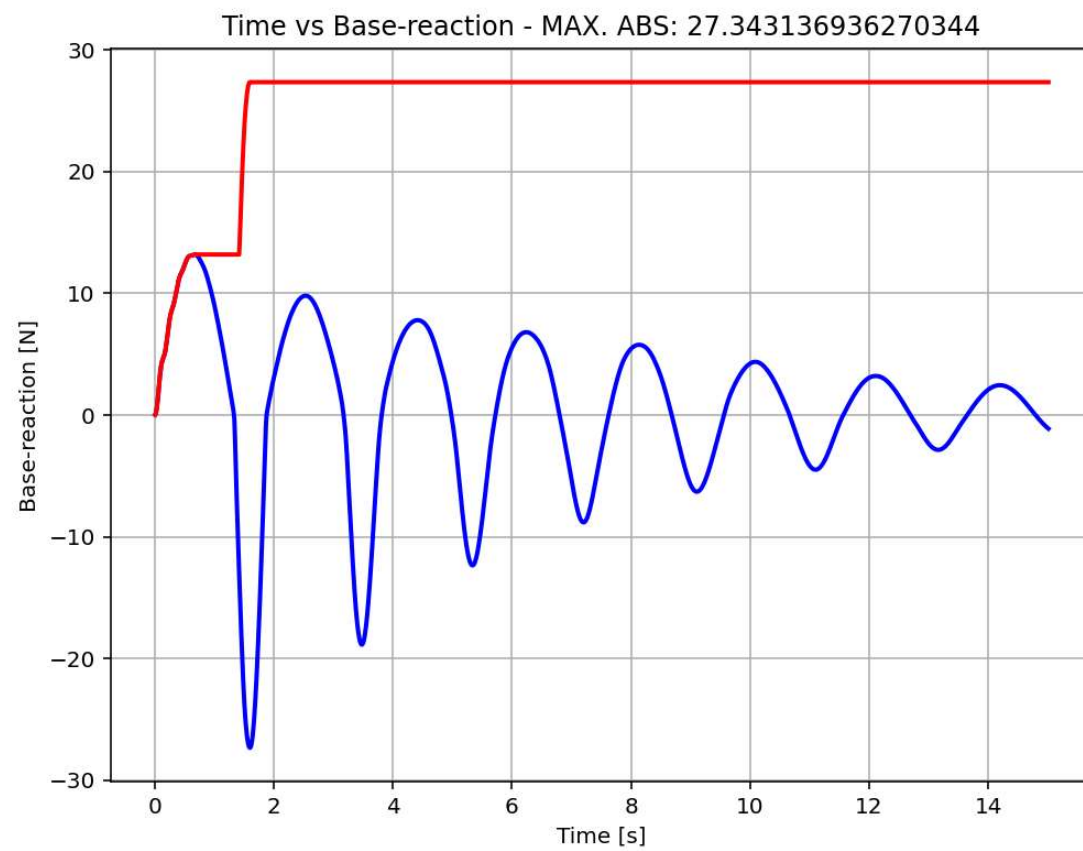
Time vs Displacement - MAX. ABS: 0.012426530254973176 | ξ (Calculated): 8.84944e+00 %

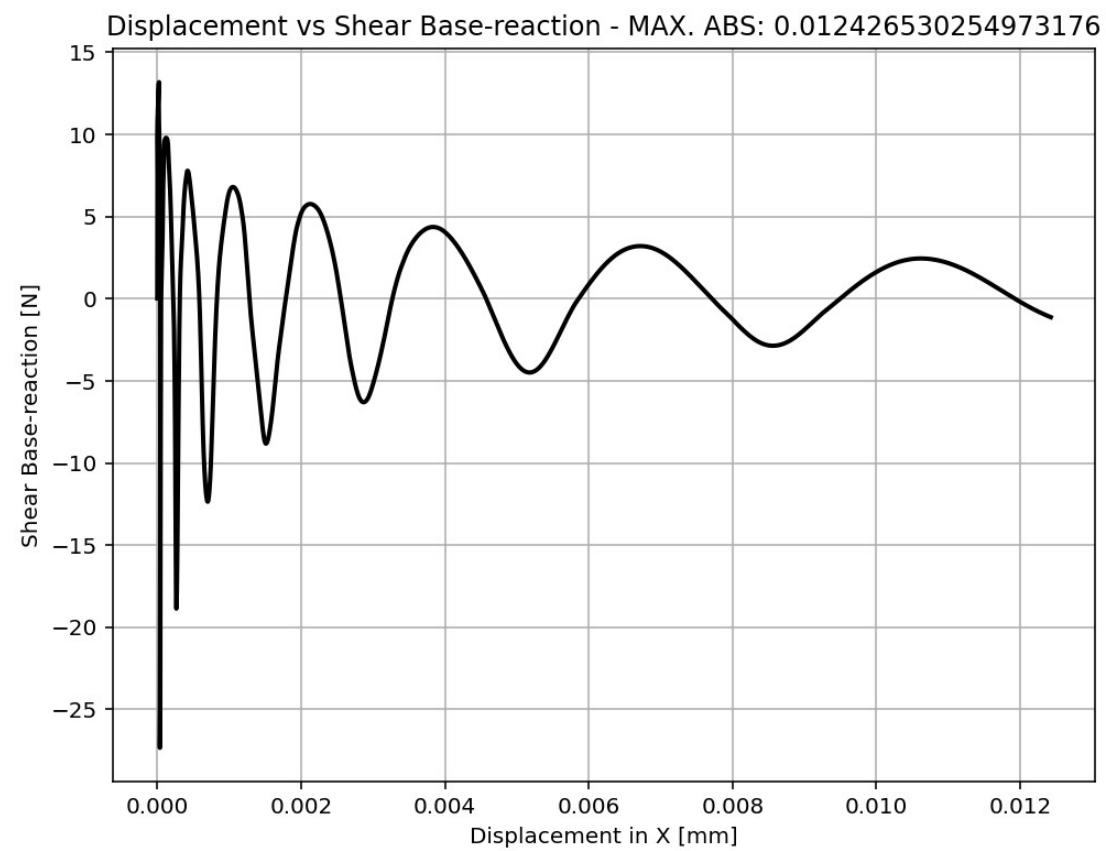












Displacement vs Axial Base-reaction - MAX. ABS: 0.0024697915505548637

